

Similarity

How similar are two nodes in a network?

Structural Equivalence

- **Structural Equivalence:**

- We look at the neighborhood shared by two nodes;
- The size of this shared neighborhood defines how similar two nodes are.

- ***Example:***

- *Two brothers have in common*
 - *sisters, mother, father, grandparents, etc.*
- *This shows that they are similar,*
- *Two **random** male or female individuals do not have much in common and are dissimilar.*

Structural Equivalence: Definitions

- **Vertex similarity:** $\sigma(v_i, v_j) = |N(v_i) \cap N(v_j)|$

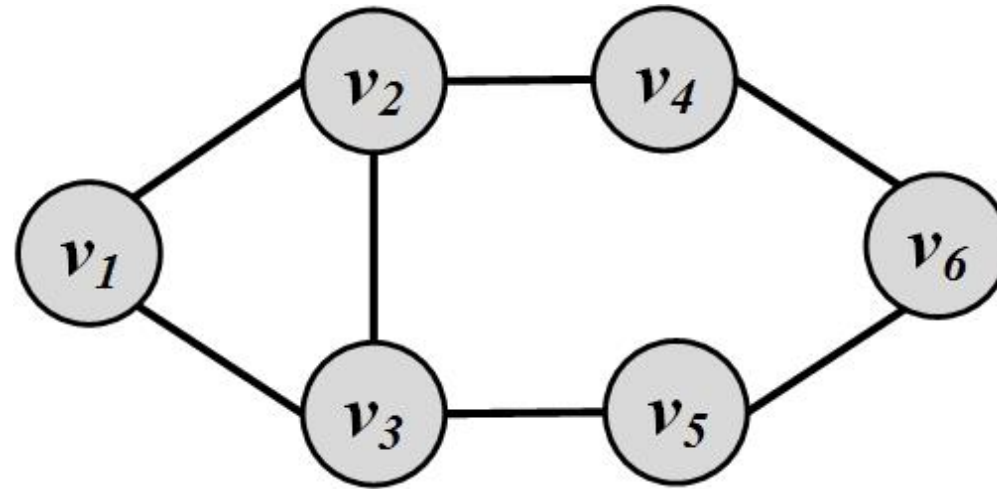
Normalize?

Jaccard Similarity: $\sigma_{Jaccard}(v_i, v_j) = \frac{|N(v_i) \cap N(v_j)|}{|N(v_i) \cup N(v_j)|}$

Cosine Similarity: $\sigma_{Cosine}(v_i, v_j) = \frac{|N(v_i) \cap N(v_j)|}{\sqrt{|N(v_i)| |N(v_j)|}}$

- The neighborhood $N(v)$ often excludes the node itself v .
 - **What can go wrong?**
 - Connected nodes not sharing a neighbor will be assigned zero similarity
 - **Solution:**
 - We can assume nodes are included in their neighborhoods

Similarity: Example



$$\sigma_{\text{Jaccard}}(v_2, v_5) = \frac{|\{v_1, v_3, v_4\} \cap \{v_3, v_6\}|}{|\{v_1, v_3, v_4, v_6\}|} = 0.25$$

$$\sigma_{\text{Cosine}}(v_2, v_5) = \frac{|\{v_1, v_3, v_4\} \cap \{v_3, v_6\}|}{\sqrt{|\{v_1, v_3, v_4\}| |\{v_3, v_6\}|}} = 0.40.$$

Measuring Similarity Significance: compare the calculated similarity value with its expected value where vertices pick their neighbors at random

- For vertices v_i and v_j with degrees d_i and d_j the expected number of common neighbors is $d_i d_j / n$
 - There is a d_i / n chance of becoming v_i 's neighbor
 - v_j selects d_j neighbors
- We can rewrite neighborhood overlap as

$$\sigma(v_i, v_j) = |N(v_i) \cap N(v_j)| = \sum_k A_{i,k} A_{j,k}$$

Normalized Similarity, cont.

$$\begin{aligned}\sigma_{\text{significance}}(v_i, v_j) &= \sum_k A_{i,k} A_{j,k} - \frac{d_i d_j}{n} & \bar{A}_i &= \frac{1}{n} \sum_k A_{i,k} \\&= \sum_k A_{i,k} A_{j,k} - n \frac{1}{n} \sum_k A_{i,k} \frac{1}{n} \sum_k A_{j,k} \\&= \sum_k A_{i,k} A_{j,k} - n \bar{A}_i \bar{A}_j \\&= \sum_k (A_{i,k} A_{j,k} - \bar{A}_i \bar{A}_j) \\&= \sum_k (A_{i,k} A_{j,k} - \bar{A}_i \bar{A}_j - \bar{A}_i \bar{A}_j + \bar{A}_i \bar{A}_j) \\&= \sum_k (A_{i,k} A_{j,k} - A_{i,k} \bar{A}_j - \bar{A}_i A_{j,k} + \bar{A}_i \bar{A}_j) \\&= \sum_k (A_{i,k} - \bar{A}_i)(A_{j,k} - \bar{A}_j)\end{aligned}$$

What is this?

Normalized Similarity, cont.

n times the Covariance between A_i and A_j

$$\frac{1}{n} \sum_k (A_{i,k} - \bar{A}_i)(A_{j,k} - \bar{A}_j)$$

Normalize covariance by the multiplication of Variances.

$$\sqrt{\frac{1}{n} \sum_k (A_{i,k} - \bar{A}_i)^2} \sqrt{\frac{1}{n} \sum_k (A_{j,k} - \bar{A}_j)^2}$$

We get **Pearson correlation coefficient**

$$\begin{aligned} \sigma_{\text{pearson}}(v_i, v_j) &= \frac{\sigma_{\text{significance}}(v_i, v_j)}{\sqrt{\sum_k (A_{i,k} - \bar{A}_i)^2} \sqrt{\sum_k (A_{j,k} - \bar{A}_j)^2}} \\ &= \frac{\sum_k (A_{i,k} - \bar{A}_i)(A_{j,k} - \bar{A}_j)}{\sqrt{\sum_k (A_{i,k} - \bar{A}_i)^2} \sqrt{\sum_k (A_{j,k} - \bar{A}_j)^2}} \end{aligned}$$

(range of $\sigma \in [-1,1]$)